

Medical Conferences

Category	Details	Metrics	Formulas	Purpose
Physician Engagement	Assess booth interaction and conversion potential	Interaction Rate: % of booth visitors interacting	$\frac{\# \text{ Physician IDs of Interactions}}{\text{Booth Traffic}}$	Understand intensity of interactions
	Explore engagement patterns and conversion trends	Time Spent Distribution: Engagement duration	25%/ 50%/ 75% time spent	Measure engagement quality
	Identify impactful interaction characteristics	Information Submission Rate: Personal info collection rate	$\frac{\# \text{ of Personal Info Submissions}}{\# \text{ Physician IDs of Interactions}}$	Evaluate physician interest
		Prescription Rate: Lead conversion	$\frac{\# \text{ of Prescriptions}}{\# \text{ Personal Info Provided}}$	Measure lead conversion success
ROI	Quantify financial effectiveness of booth participation	3-Month ROI: Short-term return	$\frac{\text{Revenue from 3-Month Prescription}}{\text{Total Cost}}$	Assess short-term ROI
	Identify areas for cost optimization	6-Month ROI: Medium-term return	$\frac{\text{Revenue from 6-Month Prescription}}{\text{Total Cost}}$	Evaluate longer-term ROI effectiveness

Executive Summary

✓ Main findings

- Seasonal Peaks: Prescription volumes peak every 2-3 months, driven by quarterly marketing efforts.
- March 2013 Impact: Record-high marketing efforts (e.g., loyalty cards, free samples) led to the highest prescription peak.
- Loyalty Cards: Most effective (+230% lift, $\beta = 2.098$, $p < 0.01$), optimal usage is 5; impact declines beyond 5.
- Free Samples & Sales Reps: Free samples (+92%, $\beta = 0.523$) and sales reps (+73%, $\beta = 0.342$) significantly boost prescriptions.
- Time Lag: Marketing actions show a 2-6 week delay before impacting prescription volumes.

✓ Recommendations

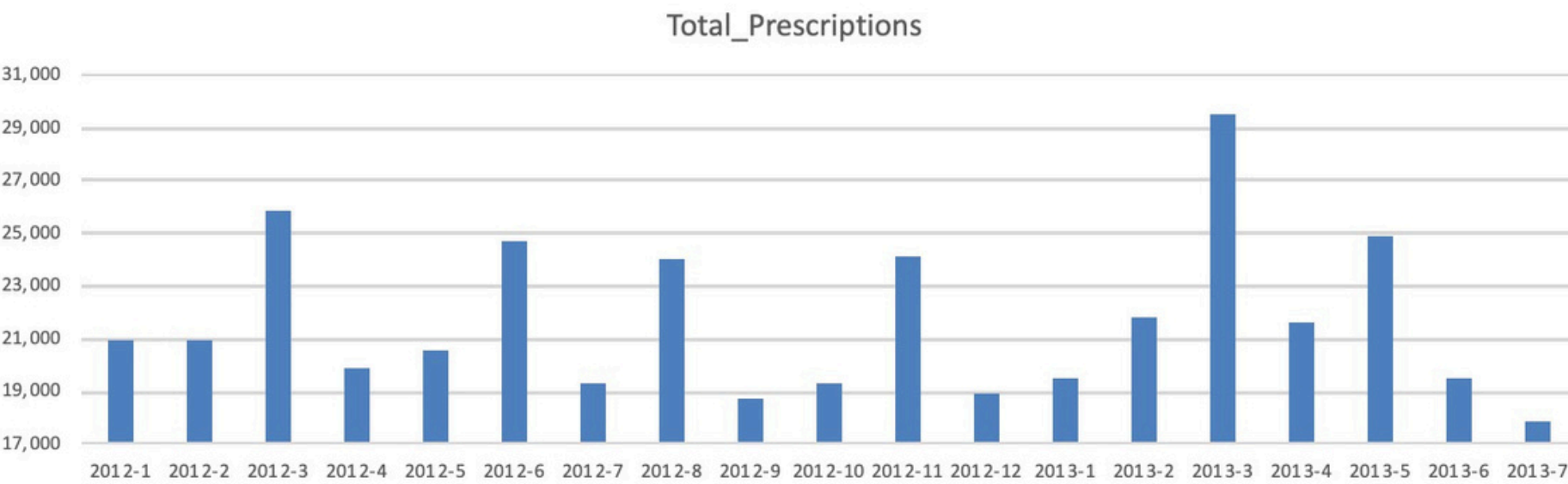
- Expand Loyalty Cards: Focus on reaching more patients in high-impact regions to drive sustained prescriptions. Prioritizing distribution of up to 6 cards per physician.
- Optimize Free Samples: Target high-potential physicians and patient segments to maximize conversions. Focus on distributing 50-75 free samples per physician.
- Refocus Speaker Programs: Shift resources to more effective initiatives or target underperforming areas.
- Leverage Sales Rep Visits: Use strategically to reinforce Loyalty Cards and Free Samples.

Seasonal Patterns:

- Peaks are observed roughly every 2-3 months (e.g.Mar 2012, Jun, Aug, Nov, Mar 2013), suggesting a recurring cycle in the effectiveness of marketing efforts.
- These peaks likely align with quarterly marketing efforts, including higher speaker attendance, more free samples distributed, and increased visits from sales representatives.

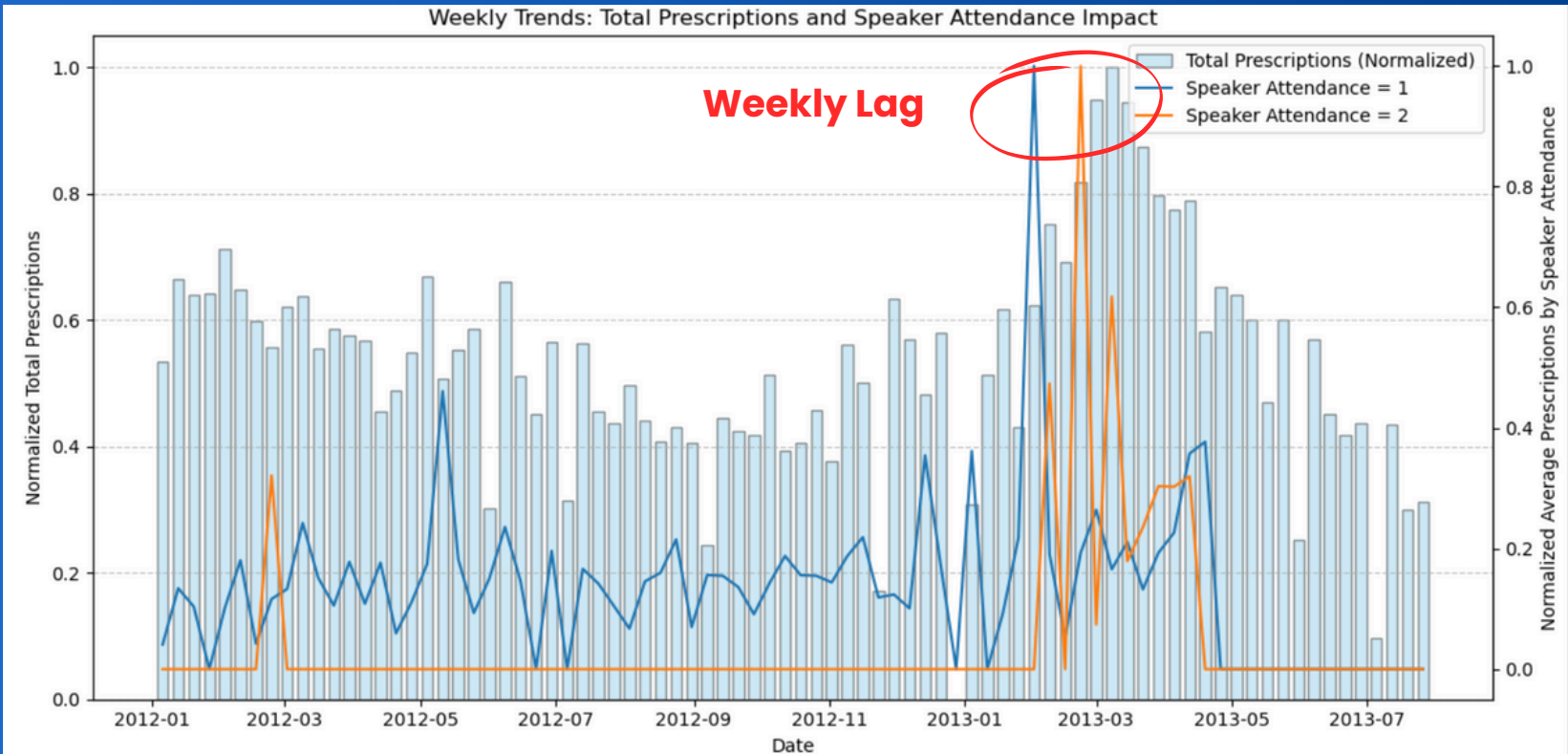
Extreme High Prescription:

- Mar 2013: The highest overall peak in Total Prescriptions aligns with record-high levels of Speaker Attendance (200), Free Samples (32,117), Loyalty Cards (904), and Sales Rep Visits (9,495), highlighting the exceptional impact of these initiatives when significantly amplified.



Speaker Attendance	44	81	131	141	67	81	53	81	98	114	80	39	31	44	200	55	0	0	0
Free Samples	26290	34047	39119	27222	30725	31242	28324	36856	25785	26764	31620	21304	27613	24109	32117	24248	35116	23929	24108
Loyalty Cards	6	6	22	29	46	70	64	63	117	266	486	445	496	612	904	768	1059	938	924
Sales Rep Visits	6006	8680	11078	8215	8931	8972	7960	10280	7777	8622	9496	6509	7183	6083	9495	7567	9291	6027	6515

Prescription Trend & Cycles



Time Lag in Weeks:

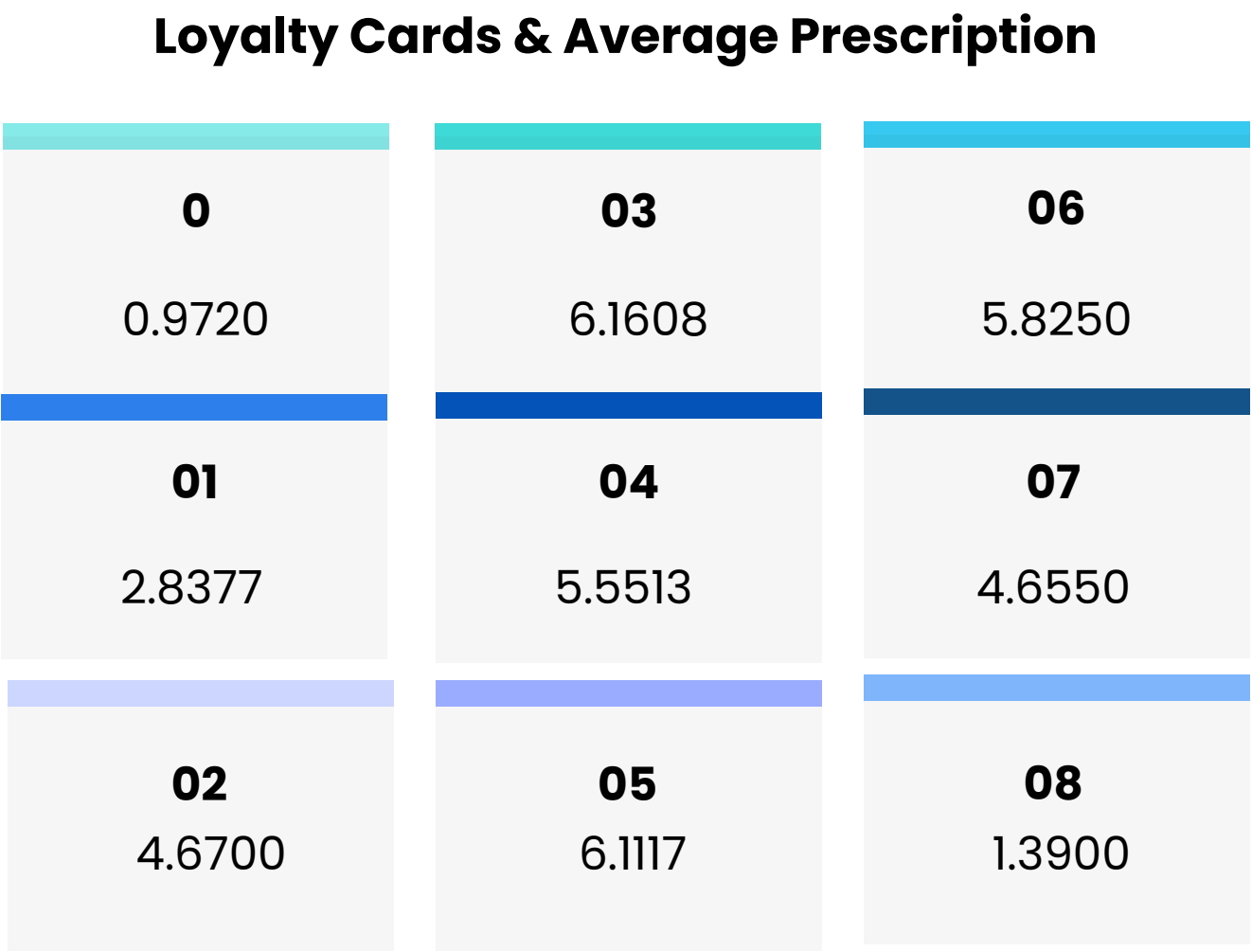
Sharp increases in Speaker Attendance closely align with subsequent prescription peaks (e.g., 2012-03 and 2013-03), suggesting a short time lag of 2-6 weeks between sample distribution and changes in prescription volumes.

	Mean Treatment	Mean Control	Lift(%)
Speaker Attendance	1.6382	1.002	63.4838
Free Sample	1.6855	0.8774	92.0992
Loyalty Cards	3.2082	0.9720	230.0522
Sales Rap Visits	1.3926	0.8022	73.6018

Speaker Programs:
Lift: Smaller relative lift (+63%) vs. other programs.
Regression: $\beta_1 = 0.413$ ($p < 0.01$), rejecting H_0 with a smaller absolute impact.

Free Samples:
Lift: Significant lift (+92%).
Regression: $\beta_2 = 0.523$ ($p < 0.01$), rejecting H_0 and indicating meaningful impact.

Sales Rep Visits:
Lift: Moderate lift (+73%).
Regression: $\beta_4 = 0.342$ ($p < 0.01$), rejecting H_0 with the smallest impact among programs.



Loyalty Cards:
Lift Analysis: Loyalty Cards generated the highest relative improvement in prescriptions (+230%).

Regression Analysis(A/B Testing):
 $\beta_3 = 2.098$ ($p < 0.01$) confirms a significant impact on prescriptions. Prescriptions peak at loyalty card usage of 5, then decline beyond this point.

Methodology

Step 1: Overall Analysis

Time Series Analysis:

Trend Analysis: Monthly prescription totals are visualized to track trends.

Activity Breakdown: Marketing activities categorized (e.g., events, samples).

Participation Metrics: Physician participation frequency is tracked monthly.

Visuals: Gradient colors
(red  = high, blue  = low)
indicate activity frequency.

Objective: Identify patterns in participation and prescription volumes for further exploration.

Step2: Subgroup Analysis

A/B Testing:

- **Groups:** Compare “Participants” (Group A) vs. “Non-Participants” (Group B).
- **Hypothesis:** Marketing efforts lead to increased prescriptions.
- **Test:** If Group B outperforms Group A, the hypothesis is rejected.

Lifting Analysis:

- **Segmentation:** Classify physicians based on activity frequency.
- **Measure:** Calculate the prescription lifting (percentage difference between treated and control groups).

Objective: Identify optimal activity levels for maximizing prescription impact.

Methodology

Program	Regression Coefficient
Speaker	0.413
Free Sample	0.523
Loyalty Card	2.098
Sales Rep Visits	0.342
Constant	0.774

$$y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 D_{4i} + \epsilon_i$$

Regression

Data Preparation: Create dummy variables for each program. Physicians who did not participate in a program were set as the control group (value = 0), and those who participated were assigned a value of 1, regardless of the intensity.

Regression: Measure the overall impact of each program on prescriptions, comparing the control and treatment groups while controlling for other factors.

Subgroup

Grouped physicians by participation levels (e.g., number of Sales Rep Visits or Free Samples distributed) and calculated metrics such as average prescriptions and total prescriptions to observe trends.

$$\text{Metric}_g = \frac{\sum_{i \in g} y_i}{N_g}$$

g: Subgroup defined by participation intensity

y_i: Number of prescriptions for physician i.

N_g: Number of physicians in subgroup g.